1. Objective

What was the task about?

In this task, I learned how to perform basic active reconnaissance using browser Developer Tools, ping, traceroute, telnet, netcat, and curl. I understood how these tools help check whether a host is reachable, examine network paths, inspect website technologies, and identify services running on specific ports.

2. Concept / Understanding

What did you learn?

I learned that active reconnaissance involves directly interacting with a target system to collect information about its network and services. I understood how browser Developer Tools can reveal website technologies, HTTP headers, JavaScript files, and certificate details. I also learned how ping checks host reachability, traceroute displays the network path, and telnet and netcat help connect to specific ports and inspect service banners. Additionally, I learned how curl can retrieve HTTP response headers.

3. Practical Work

What did you perform? Write only what you actually did.

1. I studied the concept of active reconnaissance and understood how it helps gather information about a target through direct interaction.

2. I opened a web browser and learned how to access Developer Tools using Ctrl + Shift + I.

3. I explored the browser's Developer Tools to understand how website technologies and page resources can be inspected.

4. I learned how to inspect HTTP request and response headers using the browser's Network panel.

5. I studied how JavaScript source files can reveal information about a website's functionality.

6. I learned how to inspect website certificate details to understand basic HTTPS and certificate information.

7. I studied the ping command to understand how it checks whether a host is reachable.

Command: ping -c 10 10.49.161.135

8. I learned the Windows syntax for sending multiple ping requests to a target.

Command: ping -n 10 10.49.161.135

9. I studied how IPv6 connectivity can be checked using the ping command.

Command: ping -6 MACHINE_IPV6

10. I learned how traceroute displays the network path between my system and a target.

Command: traceroute 10.49.161.135

11. I studied the Windows alternative for tracing a network path.

Command: tracert 10.49.161.135

12. I learned how to trace routes to IPv6 destinations.

Command: traceroute -6 MACHINE_IPV6

13. I studied mtr and understood how it combines route tracing with real-time network path monitoring.

Command: mtr 10.49.161.135

14. I learned how telnet can establish a connection to a specific port for basic service inspection.

Command: telnet 10.49.161.135 PORT_NUMBER

15. I studied how to use netcat as a client to connect to a specific port.

Command: nc 10.49.161.135 PORT_NUMBER

16. I learned how netcat can listen on a local port for network connections.

Command: nc -lvnp PORT_NUMBER

17. I studied how netcat supports connections to IPv6 addresses.

Command: nc -6 MACHINE_IPV6 PORT_NUMBER

18. I learned how to use curl to retrieve HTTP response headers and inspect information returned by a web server.

Command: curl -I http://10.49.161.135

19. I studied how HTTPS response headers can be checked using curl.

Command: curl -I https://10.49.161.135

20. I reviewed how combining browser inspection, host reachability checks, route tracing, port connections, and HTTP header analysis provides a basic understanding of a target's network and web services.

4. Tools Used

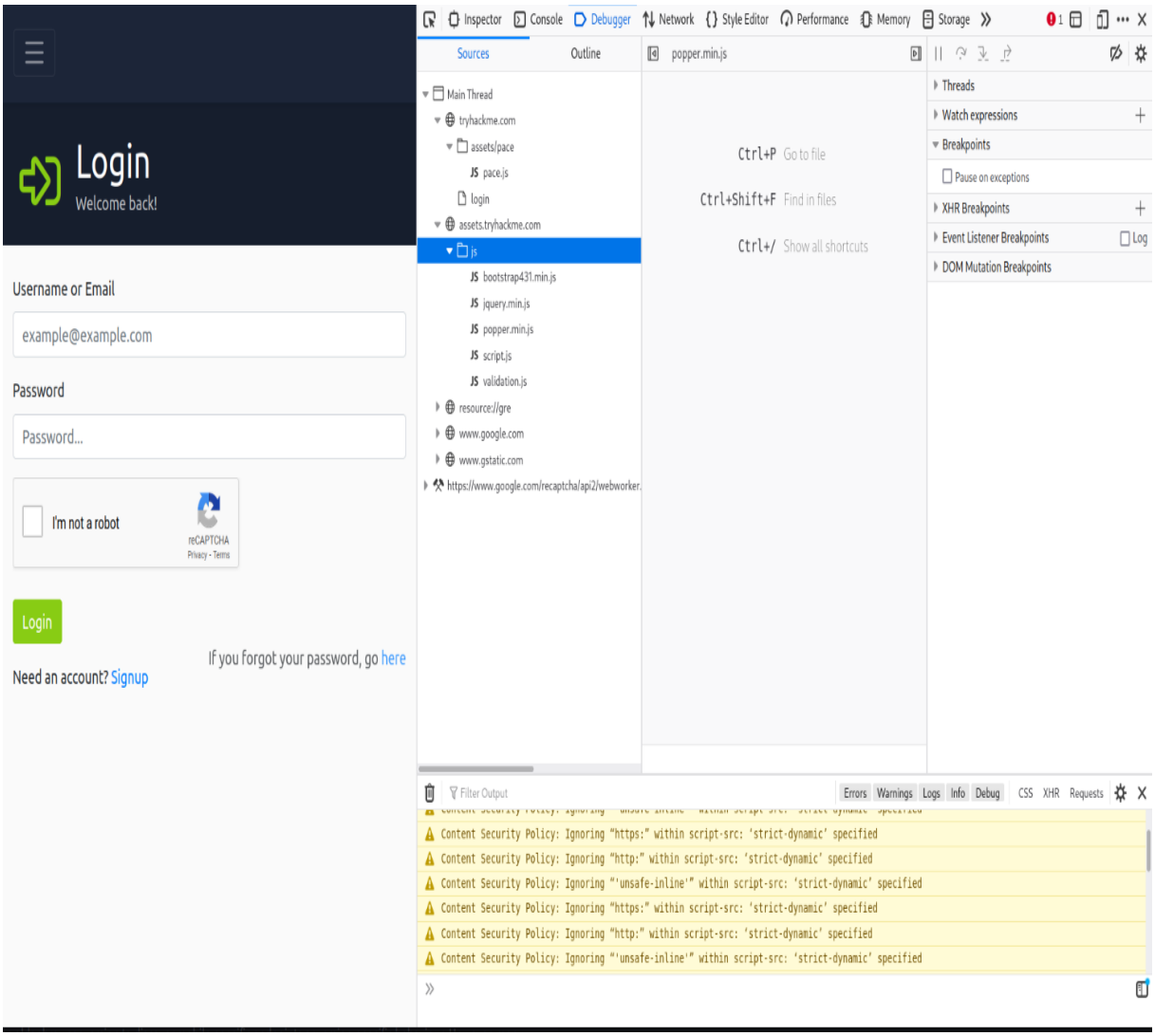
Tools used during the task.

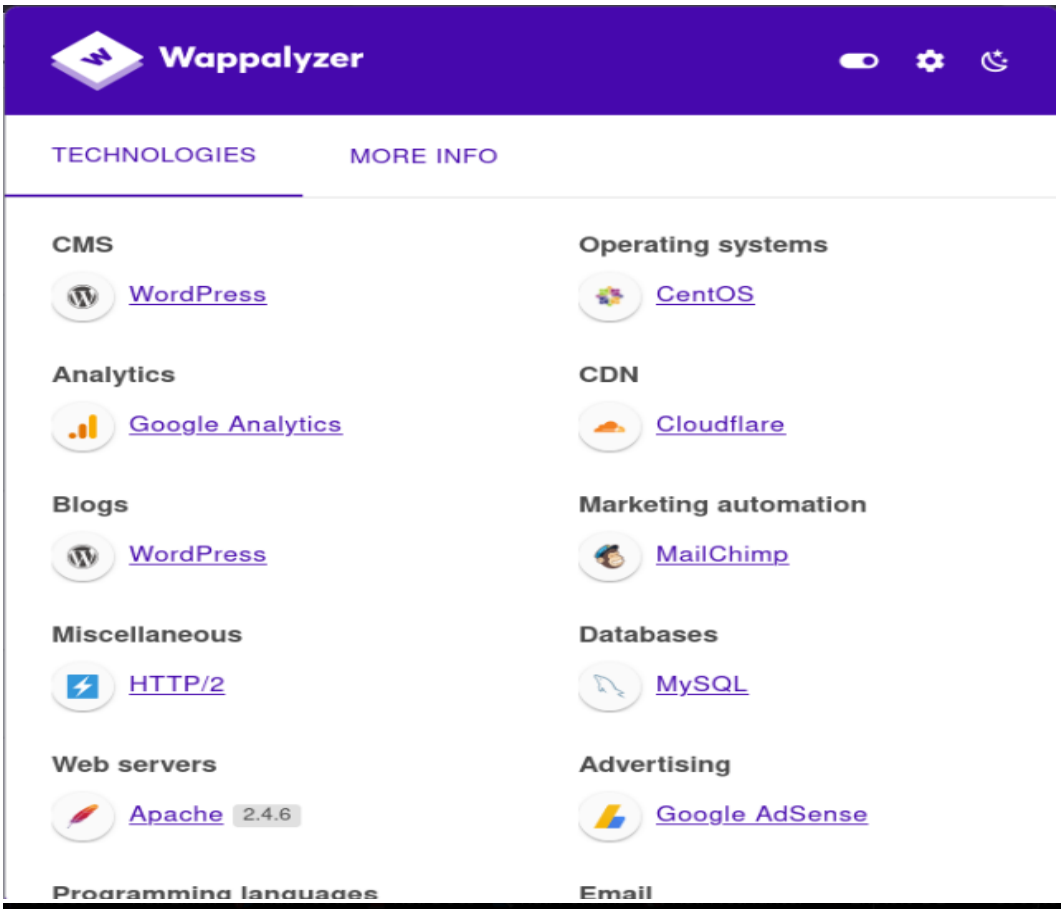
Tool	Purpose
Web Browser	To access and inspect web applications.
Browser Developer Tools	To examine technologies, HTTP headers, JavaScript files, and certificate details.
ping	To check host reachability.

Tool	Purpose
traceroute / tracert	To inspect the network path to a target.
mtr	To monitor the network path in real time.
telnet	To connect to a specific port for basic service inspection.
netcat (nc)	To connect to ports, inspect services, and listen for connections.
curl	To retrieve HTTP response headers.

5. Evidence

Relevant screenshots and results. Real screenshots only.






```
┌──(kali㉿kali)-[~]
└─$ ping -c 5 10.49.161.135
PING 10.49.161.135 (10.49.161.135) 56(84) bytes of data.
64 bytes from 10.49.161.135: icmp_seq=1 ttl=62 time=30.4 ms
64 bytes from 10.49.161.135: icmp_seq=2 ttl=62 time=30.3 ms
64 bytes from 10.49.161.135: icmp_seq=3 ttl=62 time=28.1 ms
64 bytes from 10.49.161.135: icmp_seq=4 ttl=62 time=32.9 ms
64 bytes from 10.49.161.135: icmp_seq=5 ttl=62 time=28.9 ms

— 10.49.161.135 ping statistics —
5 packets transmitted, 5 received, 0% packet loss, time 4003ms
rtt min/avg/max/mdev = 28.050/30.132/32.943/1.662 ms

┌──(kali㉿kali)-[~]
└─$ ping -c 5 tryhackme.com
PING tryhackme.com (64.239.109.129) 56(84) bytes of data.
64 bytes from 64.239.109.129: icmp_seq=1 ttl=128 time=146 ms
64 bytes from 64.239.109.129: icmp_seq=2 ttl=128 time=53.5 ms
64 bytes from 64.239.109.129: icmp_seq=4 ttl=128 time=30.0 ms
64 bytes from 64.239.109.129: icmp_seq=5 ttl=128 time=28.8 ms

— tryhackme.com ping statistics —
5 packets transmitted, 4 received, 20% packet loss, time 4004ms
rtt min/avg/max/mdev = 28.815/64.625/146.230/48.130 ms
```

```
root@ip-10-49-165-61:~# traceroute tryhackme.com
traceroute to tryhackme.com (10.49.90.216), 30 hops max, 60 byte packets
 1  * * *
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
```

```
root@ip-10-49-165-61:~# telnet 10.49.161.135 80
Trying 10.49.161.135...
Connected to 10.49.161.135.
Escape character is '^]'.
GET / HTTP/1.1
host: telnet

HTTP/1.1 200 OK
Date: Fri, 16 Oct 2026 07:40:24 GMT
Server: Apache/2.4.61 (Debian)
Last-Modified: Mon, 30 Aug 2021 12:09:24 GMT
ETag: "15-5cac5b436ddfa"
Accept-Ranges: bytes
Content-Length: 21
Content-Type: text/html

Telnet to port 80...
Connection closed by foreign host.
```

```
GET / HTTP/1.1
host: netcat

HTTP/1.1 200 OK
Server: nginx/1.6.2
Date: Tue, 17 Aug 2021 11:39:49 GMT
Content-Type: text/html
Content-Length: 867
Last-Modified: Tue, 17 Aug 2021 11:12:16 GMT
Connection: keep-alive
ETag: "611b9990-363"
Accept-Ranges: bytes
```

6. Result

What did you achieve or discover?

I learned the fundamentals of active reconnaissance and understood how different tools help examine host reachability, network paths, website information, HTTP headers, and services running on specific ports. I also learned the basic command syntax for ping, traceroute, telnet, netcat, and curl.

7. Key Learnings

What did you learn from the task?

- I learned the difference between passive and active reconnaissance.
- I understood how Browser Developer Tools help inspect website technologies and resources.
- I learned how to use ping to check host reachability.
- I understood how traceroute, tracert, and mtr help examine network paths.
- I learned how telnet and netcat connect to specific ports.
- I understood how service banners can provide information about running services and their versions.
- I learned how curl -I retrieves HTTP response headers.
- I understood the importance of using these tools only on authorised targets and within the permitted testing scope.